Mock Interview: Dynamo

Test your knowledge of Dynamo's architecture by answering these questions.

Quiz ?

Statement

How does Dynamo scale?

Answer

Dynamo uses **consistent hashing** to scale. Furthermore, with the help of Vnodes, adding nodes to the cluster is quite easy. For distributing the load, any newly added node receives Vnodes from existing nodes.

Hide Answer

Quiz ?

Statement

How does Dynamo ensure fault-tolerance and reliability of data?

Answer

Dynamo replicates data to multiple nodes for reliability and fault-tolerance.

Hide Answer

Quiz ?

Statement

How does Dynamo provide an always writable experience?

Answer

While writing data, if the required node is unreachable, another node can accept writes on its behalf. The write is then kept in a local buffer and sent out once the destination node is healthy again. This process is called **Hinted handoff**, and it makes Dynamo "always writeable" even in the extreme case where only a single node is alive, write requests will still get accepted and eventually processed.

Hide Answer

Quiz ?

Statement

How does Dynamo ensure durability?

Answer

Dynamo ensures durability by replicating data across multiple replicas, as well as storing them on the disk.

Hide Answer

Quiz ?

Statement

How can Dynamo be configured for faster reads or faster writes?

Answer

Dynamo uses a consistency protocol similar to quorum systems. If R is the minimum number of nodes that must participate in a successful read operation. And W is the same as R, except it's for a write operation. Then $R+W > N$ yields a quorum-like system, where N is the total number of replicas. If $N=3$, we can have the following configuration. For faster writes, we can use $R=3, W=1$. This also means that reads will be slower, and data is not very durable. For faster reads, we can use $R=1, W=3$. This also means that writes will be slower, but data will be durable

Hide Answer

Quiz ?

Statement

How does Dynamo manage high availability?

Answer

Using **sloppy quorum** and **hinted handoff**, Dynamo is always available for writes even if the first N nodes in the preference list are not healthy.

Hide Answer

Quiz ?

Statement

How does Dynamo perform anti-entropy?

Answer

Dynamo uses **Merkle trees** to compare replicas of a data range. The principal advantage of using the Merkle tree is that each branch of the tree can be compared independently without requiring nodes to download the entire tree or the whole data set. Merkle trees minimize the amount of data that needs to be transferred for synchronization and reduce the number of disk reads performed during the anti-entropy process.

Hide Answer

Quiz ?

Statement

Given a key, how do nodes find out which node is responsible for managing that key?

Answer

Dynamo uses **gossip protocol** which enables each node to keep track of state information about the other nodes in the cluster, like which nodes are reachable, what key ranges they are responsible for, etc. Nodes share state information with each other to stay in-sync. Gossip protocol is a peer-to-peer communication mechanism in which nodes periodically exchange state information about themselves and about other nodes they know about. Each node initiates a gossip round every second to exchange state information about itself and other nodes with one other random node. This means that any new events will eventually propagate through the system, and all nodes quickly learn about all other nodes in a cluster.

Hide Answer

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